The Data Lifecycle

#harvarddatafest

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Data

Facts that can be analyzed or used in an effort to gain knowledge or make decisions; information.

Latin Datum “Something Given”

(The American Heritage Dictionary)

Museum of Fine Arts
The Art of the Ancient World

Friday, January 13, 2017, 7pm

Giza Excavation 1902-1947

THE HARVARD UNIVERSITY BOSTON
MUSEUM OF FINE ARTS EXPEDITION

Archeological Finds as Data

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Photo ID number: B5789_NS

Photo subjects: Object(s) photograph: Site: Giza; view: G 7632, G 7510, street G 7000, street G 7300, street G 7400, street G 7500, avenue G 2

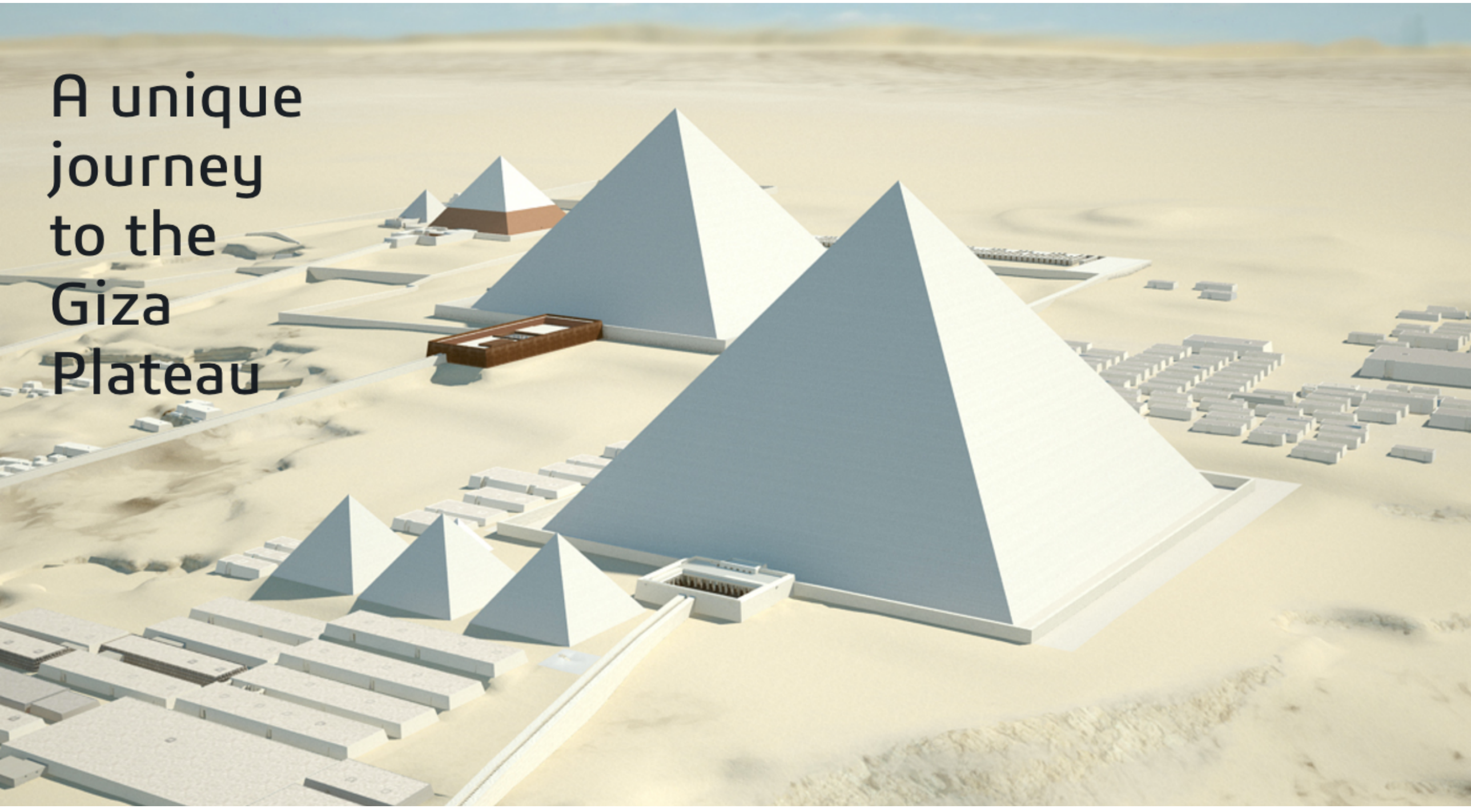
Description: Bronze objects: top row: [G 7632 A II] 25-2-822 (= MFA 25.2436, amulet of Horus), [G 7510 G] 25-2-916a (= MFA 25.2465, amulet of ichneumon), [street G 7500] 25-2-296 (= MFA 25.2211, Apis bull amulet), [street G 7300] 25-1-143 (= MFA 25.1591, Apis bull amulet), [avenue G 2] 25-1-1363 (= MFA 25.2066, Apis bull amulet), [street G 7500] 25-2-297 (= MFA 25.2212, amulet of Anubis); second row, left: [street G 7500] 25-1-1385 (= MFA 25.2082, ladle); second row, right: [G 7130-7140: G 7130 A] 24-12-513 (= MFA 24.2866, unidentified deity, fragment), [avenue G 2] 25-1-1364 (= MFA 25.2067, fragment of ring); third row, right: [street G 7000] 24-11-39 (= MFA 24.2271, fish on standard); bottom row: [street G 7000] 24-11-838 (= MFA 24.2638, cat on standard), [G 7510 D] 25-3-252 (= MFA 25.2672, flail), [street G 7400] 25-1-620 (= MFA 25.1766, beard), [street G 7000] 24-11-223 (= MFA 24.2371, amulet of Mehit and Inher), [avenue G 2] 25-1-1424 (= MFA 25.2103, uraeus), [street G 7400] 25-1-842 (= MFA 25.1853, uraeus)

Photo date: 10/21/1925

The Giza Archive and catalogue:

- 21,000 finds
- 3,000 photos
- 3000 tomb and monument records
- 3,000 diary pages
- 7,000 maps
- books, manuscripts
- metadata

3D Visualization to Re-Explore Giza



Peter Der Manuelian, Harvard's Visualization Center (Geological Museum), DARTH, and Dassault Systèmes in Paris

Books as Data

googlebooks-eng-us-all-totalcounts-20090715.tx

```
Total counts of 1grams for the eng-us-all corpus. 2009/07/15. The f
tabs, correspond to the year, the number of 1grams, the number of p
books.
1584    382341  926     1
1590    270386  581     1
1592    50762   129     1
1596    187852  721     1
1600    24755   75      1
1610    1732    16      1
1620    68848   292     1
1628    81026   228     1
1635    47119   130     1
1647    34008   110     1
1648    309670  724     1
1649    13677   59      1
1656    96789   473     3
1658    694132  978     2
1660    110823  231     1
1661    56308   187     1
1662    100701  200     1
1663    267812  574     3
1664    108710  222     2
1665    117171  236     1
1666    282982  638     3
1667    220639  439     2
1668    92431   181     1
```

```
6'10_ADJ        1912    1       1
6'10_ADJ        1964    2       2
6'10_ADJ        1966    2       2
6'10_ADJ        1970    1       1
6'10_ADJ        1971    1       1
6'10_ADJ        1972    1       1
6'10_ADJ        1977    1       1
6'10_ADJ        1978    1       1
6'10_ADJ        1980    1       1
6'10_ADJ        1981    1       1
6'10_ADJ        1983    5       5
6'10_ADJ        1984    2       2
6'10_ADJ        1986    3       2
6'10_ADJ        1987    1       1
6'10_ADJ        1991    1       1
6'10_ADJ        1992    2       2
6'10_ADJ        1993    2       2
6'10_ADJ        1994    2       2
6'10_ADJ        1995    2       2
6'10_ADJ        1996    1       1
6'10_ADJ        1999    2       2
6'10_ADJ        2002    1       1
6'10_ADJ        2003    5       5
6'10_ADJ        2004    3       3
6'10_ADJ        2006    4       4
6'10_ADJ        2008    1       1
```

Harvard Cultural Observatory (Aiden, Michel, SEAS) + Google:

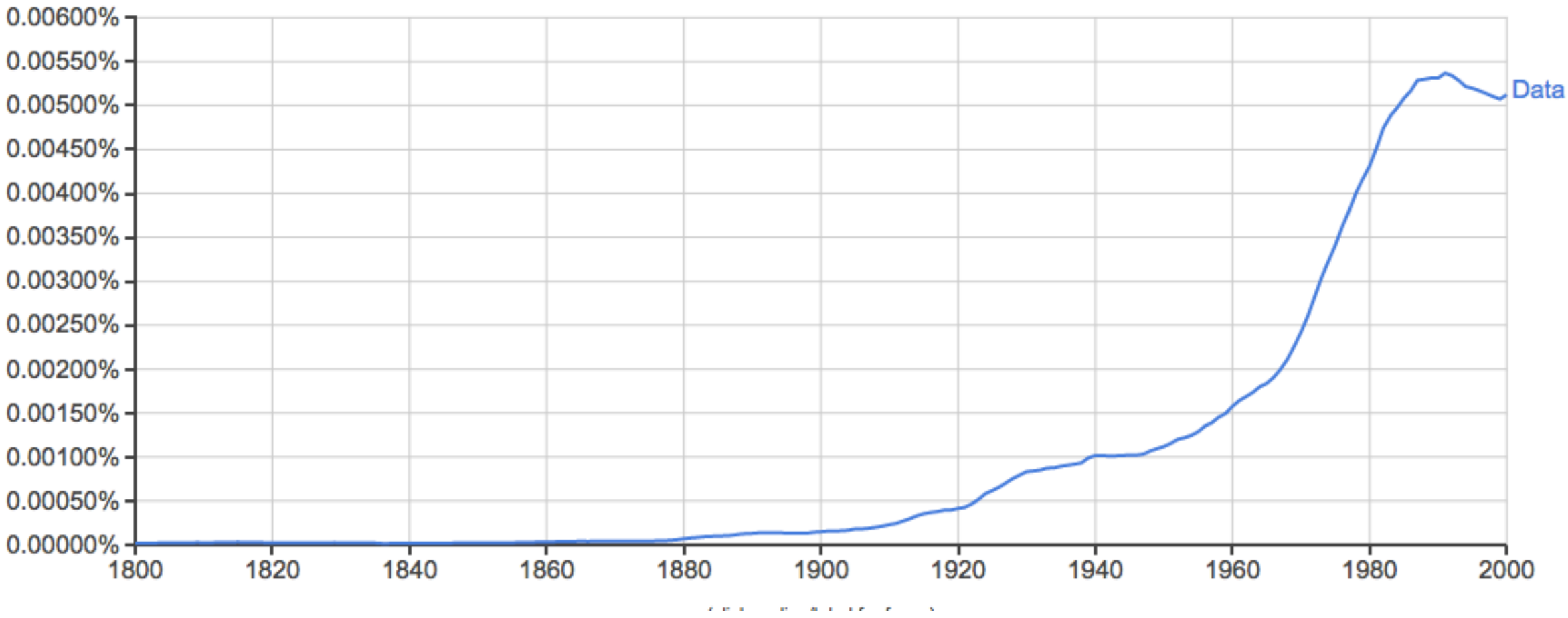
- 5 million digitized books (from130 million total)
- From year1500 to 2008
- Optical Character Recognition (OCR)
- Quantify unstructured text to n-gram count matrix

Frequency of Words with Time

Google Books Ngram Viewer

Graph these comma-separated phrases: Data case-insensitive

between 1800 and 2000 from the corpus English with smoothing of 3. Search lots of books



Historical events as Data

my ab	POSITION	Places Notes	Terrain Notes	ID	TYPE	NAME	DATE	STEP	VALUE	UNITS	TEXT	LAT1	LON1	LAT2	LON2	PATH	CERTAINTY
		Needs static event map	Needs static event map				4/6/1760	0									
							4/7/1760	**0**									
	18.3648, -76.8852	Frontier misplaced to NW		1	Conspiracy	Frontier	4/7/1760	1	5	Rebel conspirators		18.3648	-76.8852				1
	18.3542, -76.8990	Trinity misplaced to NW		1	Rebels	Trinity	4/7/1760	2	100	Rebel force		18.3542	-76.899				1
							4/8/1760	**0**									
		Still glitchy; start															

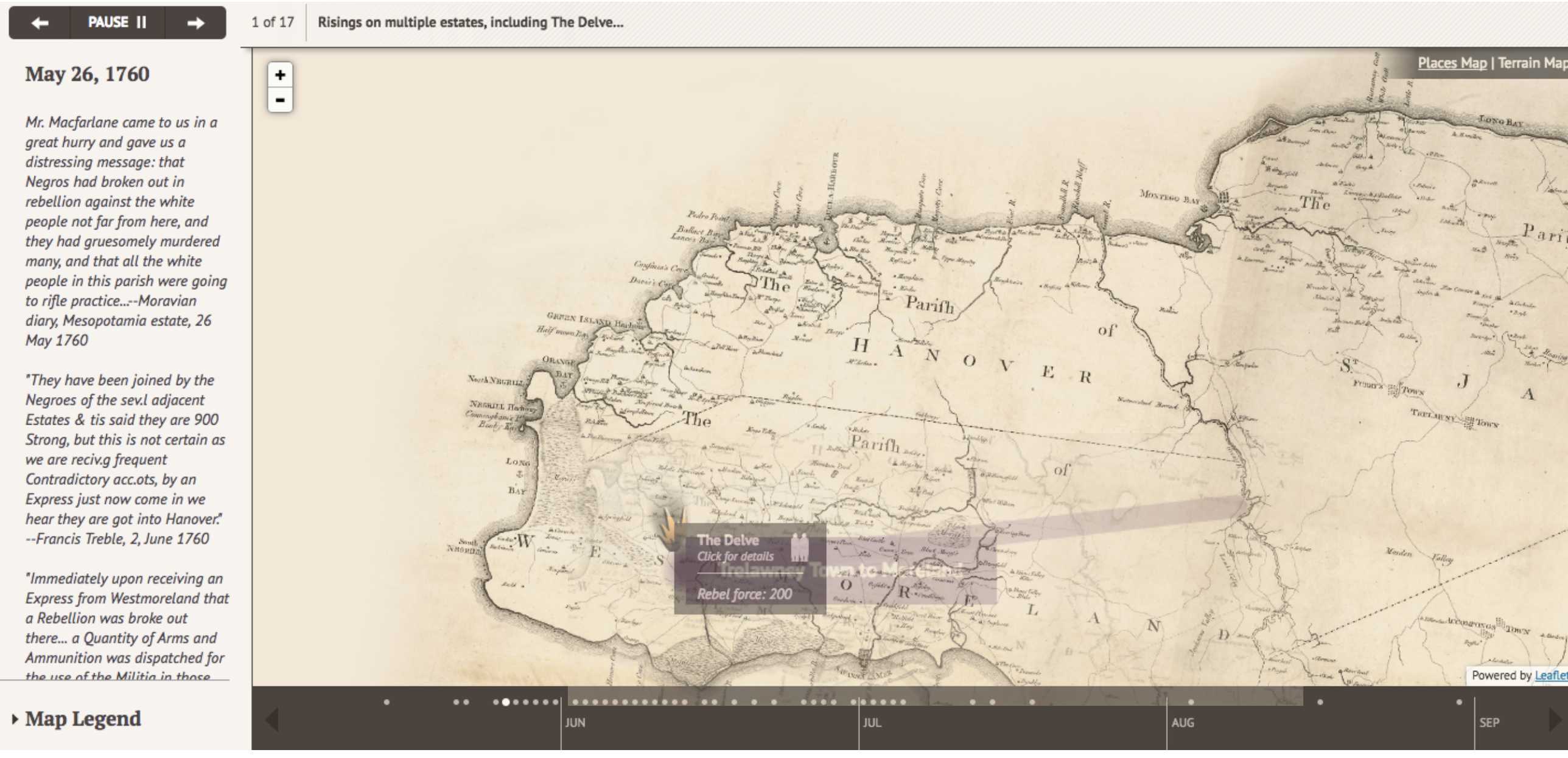
POSITION	Places Notes	Terrain Notes	ID	TYPE	NAME	DATE	S
	Needs static event map	Needs static event map				4/6/1760	
						4/7/1760	
18.3648, -76.8852	Frontier misplaced to NW		1	Conspiracy	Frontier	4/7/1760	
18.3542, -76.8990	Trinity misplaced to NW		1	Rebels	Trinity	4/7/1760	
						4/8/1760	
18.3542, -76.8990 to Ft. Haldane	**Still glitchy; start should be from Trinity**		**1**	**Rebels**	**Ft. Haldane**	**4/8/1760**	
	database fixed?		1	Clash	Ft. Haldane	4/8/1760	
18.3542, -76.8990	Trinity misplaced to NW		1	Rebels	Trinity	4/8/1760	

18.3223, -76.8875 to Esher	**Rebels should move from Ballard's Valley to Esher**	**To extent possible, movement should follow roads (dotted line)**	**1**	**Rebels**	**Esher**	**4/8/1760**	**6**			**18.3223**	**-76.8875**	**18.2809**	**-76.8675**	**-76.8882], [18.2709, -76.8851], [18.2704, -76.8822], [18.2703, -76.8773], [18.2722, -76.8736], [18.2783, -76.8703], [18.2809, -76.8675]]**	**1**
		To extent possible, movement should follow roads (dotted line)	1	Clash	Esher	4/8/1760	7		5 whites killed	18.2809	-76.8675					1
18.3223, -76.8875	**Eliminate movement; Militia appear at single point, Ballard's Valley**	**To extent possible, movement should follow roads (dotted line)**	**2**	**Militia**	**Ballard's Valley**	**4/8/1760**	**8**			**18.3223**	**-76.8875**						**0**
Esher to 18.3050, -76.8882	**Starting pt is Esher**	**To extent possible, movement should follow roads (dotted line)**	**1**	**Rebels**	**Whitehall**	**4/8/1760**	**9**	**400**	**Rebel force**	18.2809	-76.8675	**18.305**	**-76.8882**	**[[18.2809, -76.8675], [18.2792, -76.8694], [18.2766, -76.8720], [18.2740, -76.8736], [18.2708, -76.8796], [18.2706, -76.8839], [18.2809, -76.8906], [18.2885, -76.8921], [18.2908, -76.8964], [18.2919, -76.8995], [18.2952, -76.8995], [18.2981, -76.8976], [18.3032, -76.8937],[18.305, -76.8882]]**	**1**

Geospatial coordinates

Time

Jamaica Slave Revolt in Space and Time (1760 - 1761)



Animated maps to narrate and explore history data
by Vincent Brown (Harvard's History Department)

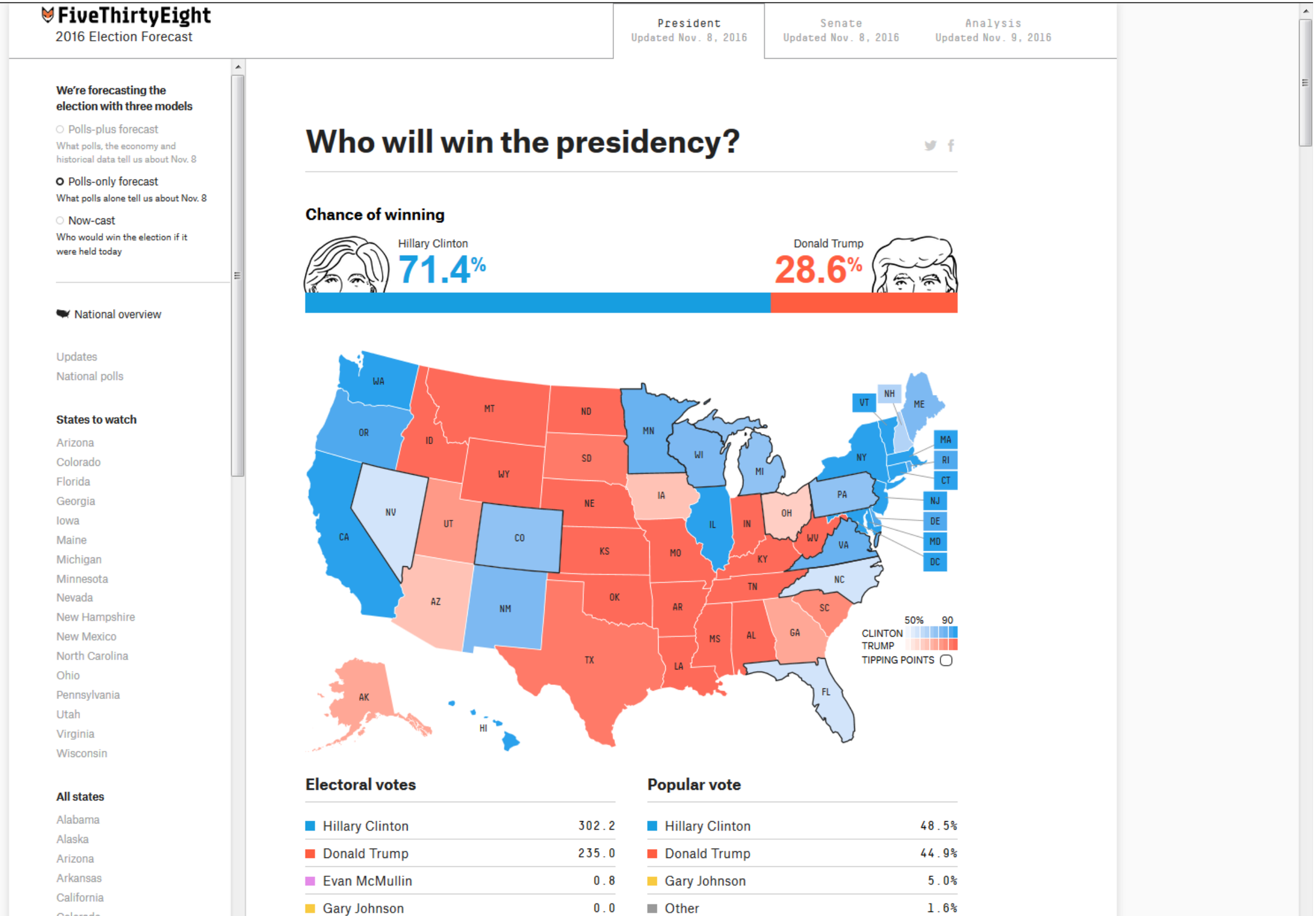
Polls as Data

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Trump	Clinton	Johnson	Other	Undecided	poll_slug	survey_hous	start_date	end_date	question_tex	sample_subp	observations	margin_of_e	mode	partisanship	partisan_affili
2	41	45	4	2	8	insights-west	Insights Wes	11/4/16	11/7/16	As you may k	Likely Voters	940	3.2	Internet	Nonpartisan	None
3	6	89	1	0	4	insights-west	Insights Wes	11/4/16	11/7/16	As you may k	Likely Voters - Democrat			Internet	Nonpartisan	None
4	82	7	3	2	6	insights-west	Insights Wes	11/4/16	11/7/16	As you may k	Likely Voters - Republican			Internet	Nonpartisan	None
5	38	43	8	4	7	insights-west	Insights Wes	11/4/16	11/7/16	As you may k	Likely Voters - independent			Internet	Nonpartisan	None
6	43	41	7	4	5	ibd-tipp-268	IBD/TIPP	11/4/16	11/7/16		Likely Voters	1107	3.1	Live Phone	Nonpartisan	None
7	42	43		5	9	ibd-tipp-268	IBD/TIPP	11/4/16	11/7/16		Likely Voters	1107	3.1	Live Phone	Nonpartisan	None
8	41	45	5	5	4	yougov-econ	YouGov/Ecor	11/4/16	11/7/16							
9	6	90	1	2	2	yougov-econ	YouGov/Ecor	11/4/16	11/7/16							
10	84	4	4	4	4	yougov-econ	YouGov/Ecor	11/4/16	11/7/16							
11	44	31	9	9	8	yougov-econ	YouGov/Ecor	11/4/16	11/7/16							
12	45	49		7		yougov-econ	YouGov/Ecor	11/4/16	11/7/16							
13	6	91		2		yougov-econ	YouGov/Ecor	11/4/16	11/7/16							
14	88	6		5		yougov-econ	YouGov/Ecor	11/4/16	11/7/16							
15	49	38		13		yougov-econ	YouGov/Ecor	11/4/16	11/7/16							
16	40	45	5	2	7	lucid-the-tim	Lucid/The Tir	11/4/16	11/6/16							
17	43	46		5	6	bloomberg-s	Bloomberg/S	11/4/16	11/6/16	If						
18	6	86		3	5	bloomberg-s	Bloomberg/S	11/4/16	11/6/16	If						
19	85	5		6	4	bloomberg-s	Bloomberg/S	11/4/16	11/6/16	If						
20	44	38		10	8	bloomberg-s	Bloomberg/S	11/4/16	11/6/16	If						
21	41	44	4	2	8	bloomberg-s	Bloomberg/S	11/4/16	11/6/16	If						
22	43	47	4	4	1	abc-post-267	ABC/Post	11/3/16	11/6/16							
23	46	49		3	2	abc-post-267	ABC/Post	11/3/16	11/6/16							
24	44	50	4	3	1	monmouth-u	Monmouth U	11/3/16	11/6/16	If						
25	44	50		5	1	monmouth-u	Monmouth U	11/3/16	11/6/16							
26	44	48	3	4	1	fox-26784	FOX	11/3/16	11/6/16							
27	4	91	2	3	0	fox-26784	FOX	11/3/16	11/6/16							
28	86	9	2	3	1	fox-26784	FOX	11/3/16	11/6/16							
29	42	36	9	9	5	fox-26784	FOX	11/3/16	11/6/16							
30	44	48		2	7	fox-26784	FOX	11/3/16	11/6/16							
31	4	90		1	4	fox-26784	FOX	11/3/16	11/6/16							
32	85	8		1	6	fox-26784	FOX	11/3/16	11/6/16							
33	43	37		3	18	fox-26784	FOX	11/3/16	11/6/16							
34	43	41	6	4	5	ibd-tipp-267	IBD/TIPP	11/3/16	11/6/16							
35	42	43		5	9	ibd-tipp-267	IBD/TIPP	11/3/16	11/6/16							
36	39	44		9	8	ipsos-reuters	Ipsos/Reuter	11/2/16	11/6/16	If						
37	7	84		5	5	ipsos-reuters	Ipsos/Reuter	11/2/16	11/6/16	If						
38	81	6		7	7	ipsos-reuters	Ipsos/Reuter	11/2/16	11/6/16	If the 2016 p	Likely Voters - Republican			Internet	Nonpartisan	None
39	24	29		28	19	ipsos-reuters	Ipsos/Reuter	11/2/16	11/6/16	If the 2016 p	Likely Voters - independent			Internet	Nonpartisan	None
40	39	42	6	6	7	ipsos-reuters	Ipsos/Reuter	11/2/16	11/6/16	If the 2016 p	Likely Voters	2195	2.4	Internet	Nonpartisan	None
41	7	81	4	3	4	ipsos-reuters	Ipsos/Reuter	11/2/16	11/6/16	If the 2016 p	Likely Voters - Democrat			Internet	Nonpartisan	None
42	81	6	5	3	5	ipsos-reuters	Ipsos/Reuter	11/2/16	11/6/16	If the 2016 p	Likely Voters - Republican			Internet	Nonpartisan	None
43	24	28	15	16	17	ipsos-reuters	Ipsos/Reuter	11/2/16	11/6/16	If the 2016 p	Likely Voters - independent			Internet	Nonpartisan	None
44	43	45	4	5	4	rasmussen-2	Rasmussen	11/2/16	11/6/16	If the 2016 p	Likely Voters	1500	2.5	IVR/Online	Nonpartisan	None
45	43	47		5	5	cbs-times-26	CBS/Times	11/2/16	11/6/16		Likely Voters	1426	3	Live Phone	Nonpartisan	None
46	7	88		1	4	cbs-times-26	CBS/Times	11/2/16	11/6/16		Likely Voters - Democrat			Live Phone	Nonpartisan	None
47	86	6		5	3	cbs-times-26	CBS/Times	11/2/16	11/6/16		Likely Voters - Republican			Live Phone	Nonpartisan	None

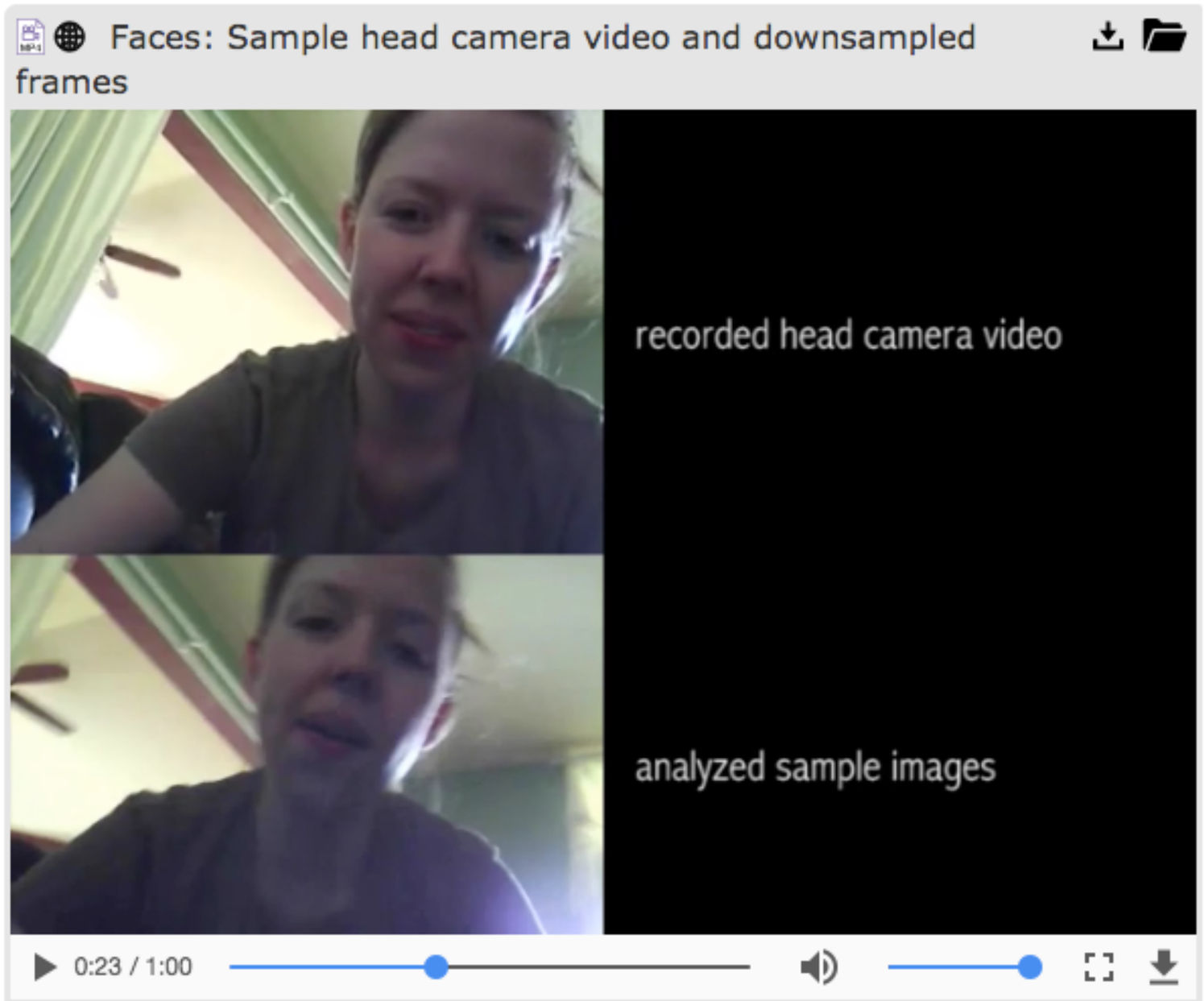
2016 Election Data:

- 100s scientific polls
- Mixed models and sample frames
  - interviewer-administered telephone polls
  - interactive-voice-response polls
  - probability-based web polls
  - good quality non-probability based web polls
- Inference and forecasting

(no comment)


FiveThirtyEight
2016 Election Forecast
President
Updated Nov. 8, 2016
Senate
Updated Nov. 8, 2016
Analysis
Updated Nov. 9, 2016
We're forecasting the election with three models
Polls-plus forecast
What polls, the economy and historical data tell us about Nov. 8
Polls-only forecast
What polls alone tell us about Nov. 8
Now-cast
Who would win the election if it were held today
National overview
Updates
National polls
States to watch
Arizona
Colorado
Florida
Georgia
Iowa
Maine
Michigan
Minnesota
Nevada
New Hampshire
New Mexico
North Carolina
Ohio
Pennsylvania
Utah
Virginia
Wisconsin
All states
Alabama
Alaska
Arizona
Arkansas
California
Who will win the presidency?
Chance of winning
Hillary Clinton
71.4%
Donald Trump
28.6%
WA
OR
CA
ID
NV
MT
WY
UT
AZ
CO
NM
ND
SD
NE
KS
OK
TX
MN
IA
MO
AR
LA
WI
IL
MS
MI
IN
KY
TN
AL
OH
WV
GA
FL
SC
NC
VA
PA
NY
VT
NH
ME
MA
RI
CT
NJ
DE
MD
DC
AK
HI
50%
90
CLINTON
TRUMP
TIPPING POINTS
Electoral votes
Hillary Clinton 302.2
Donald Trump 235.0
Evan McMullin 0.8
Gary Johnson 0.0
Popular vote
Hillary Clinton 48.5%
Donald Trump 44.9%
Gary Johnson 5.0%
Other 1.6%

Video as Data

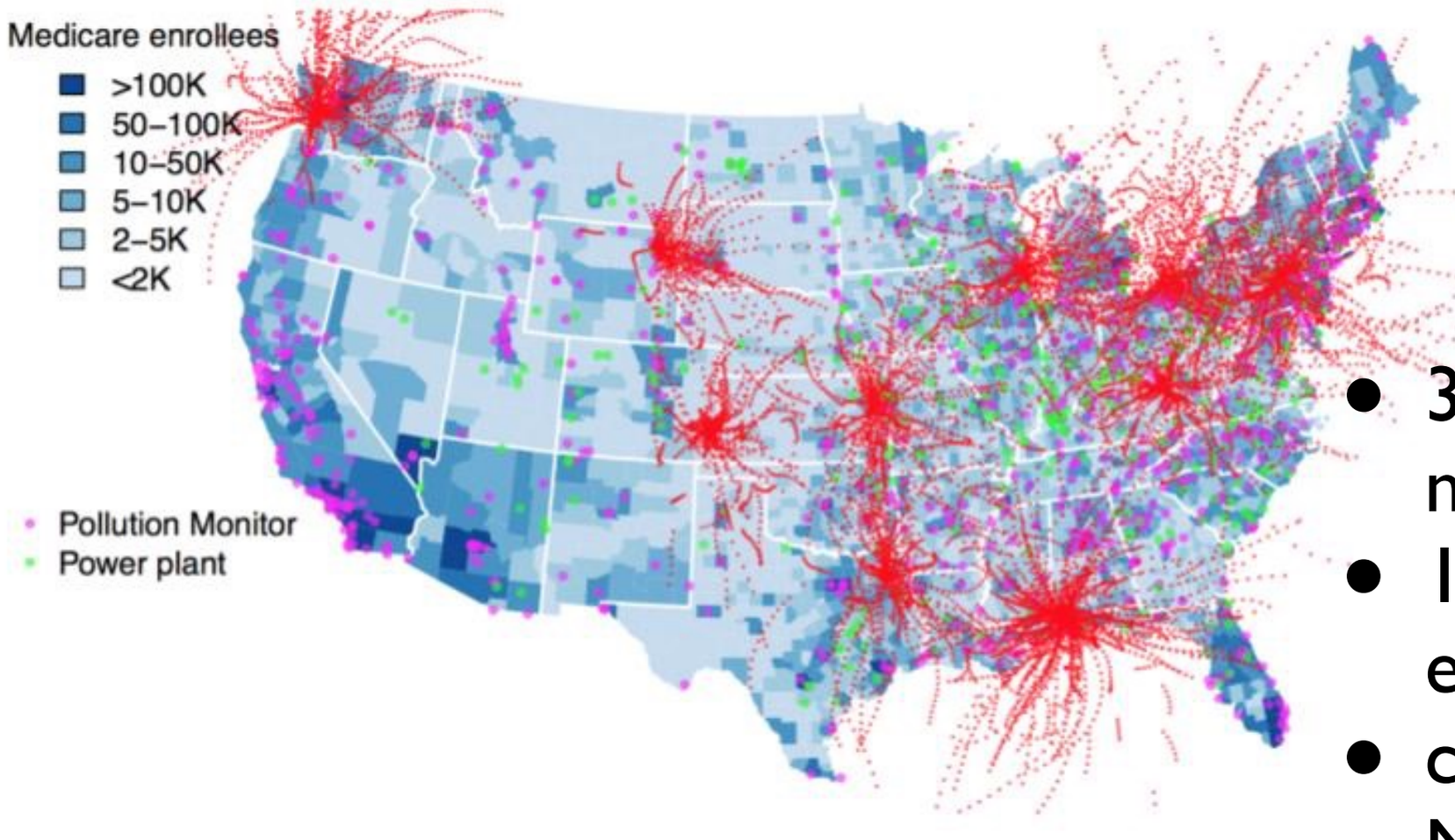


Video in developmental psychology:

- 143 hours of infant-perspective scenes, collected from 34 infants aged 1 month to 2 years
- Coding and annotations

Fausey, C.M., Smith, L.B. & Jayaraman, S. (2015). From faces to hands: Changing visual input in the first two years. Databrary

Pollution measurements as Data



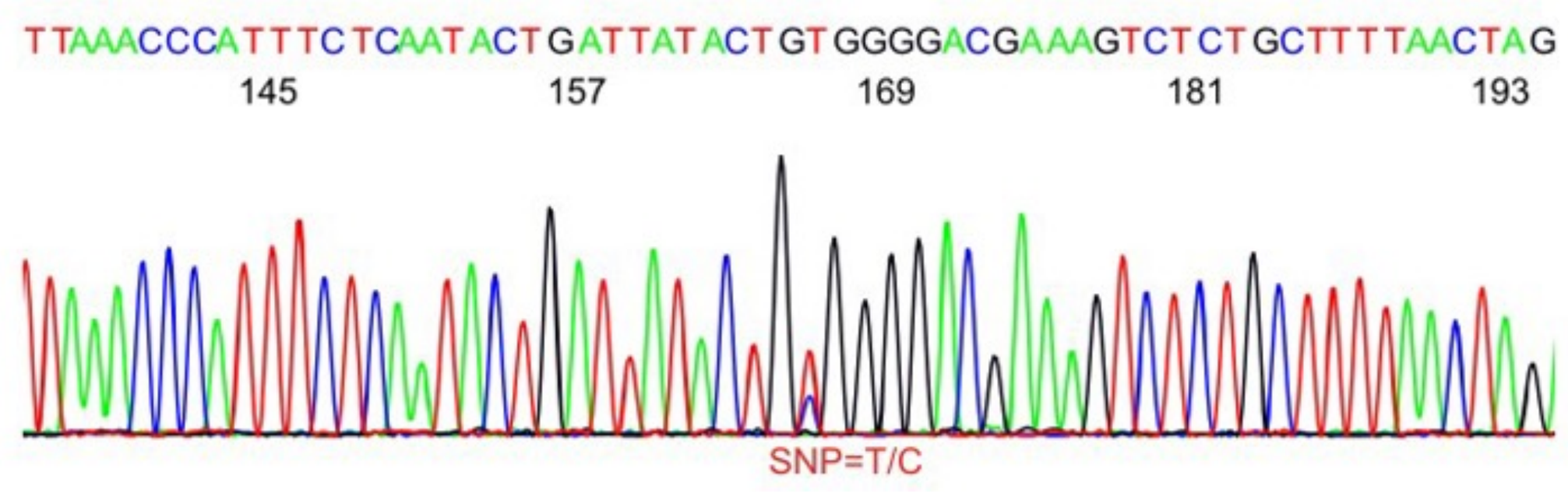
Combine data from diverse sources:

- 3000 air pollution monitors
- 1800 power plants SO_2 emissions
- claims data from 60 M Medicare beneficiaries
- air pollution trajectories based on weather models

Zigler, Dominici, Wang, Kim, Choirat (Harvard Chan School of Public Health)

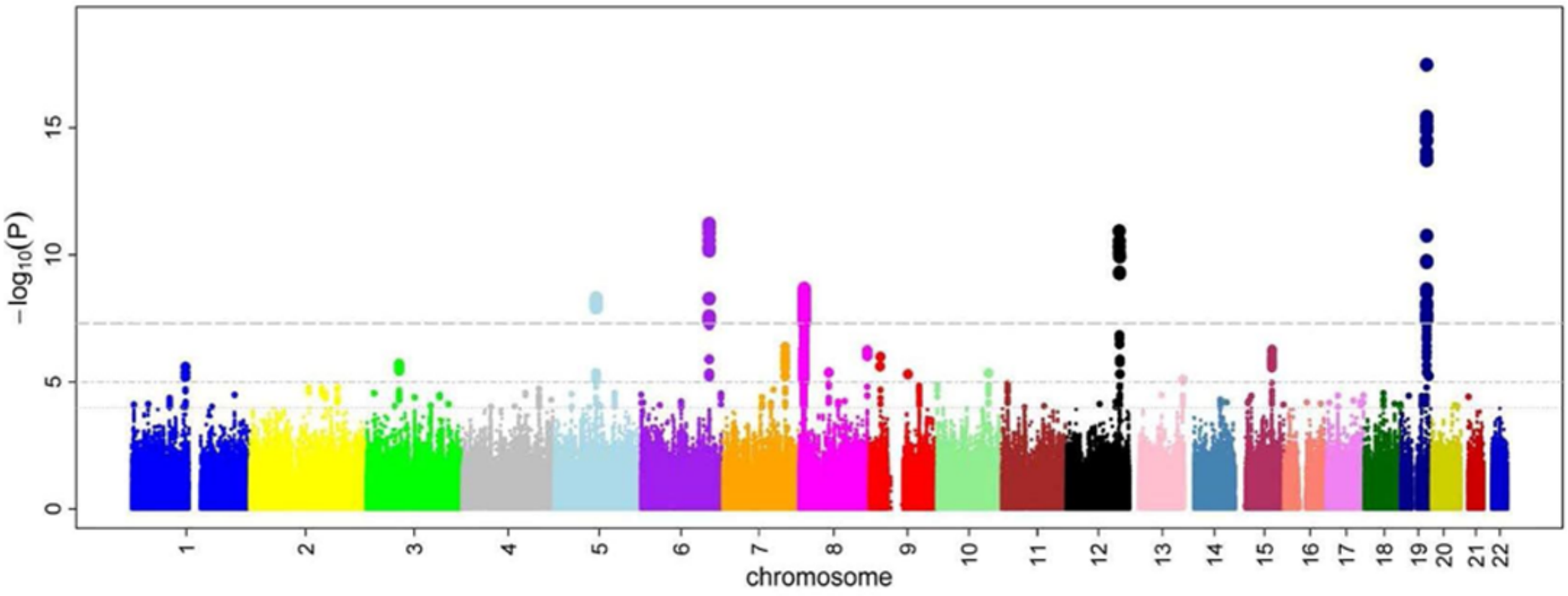
Your Genome as Data

Measurement of Single-Nucleotide Polymorphisms (SNPs) with Sequencer instrument



Genome Wide Association Study

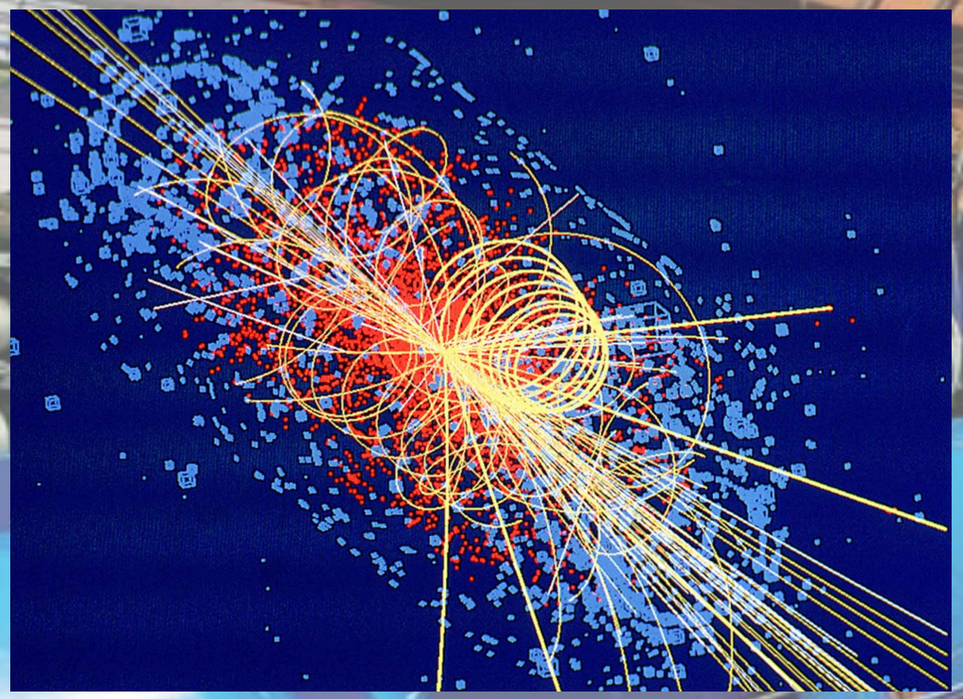
Comparing genomes from individuals with (cases) and without disease (controls)

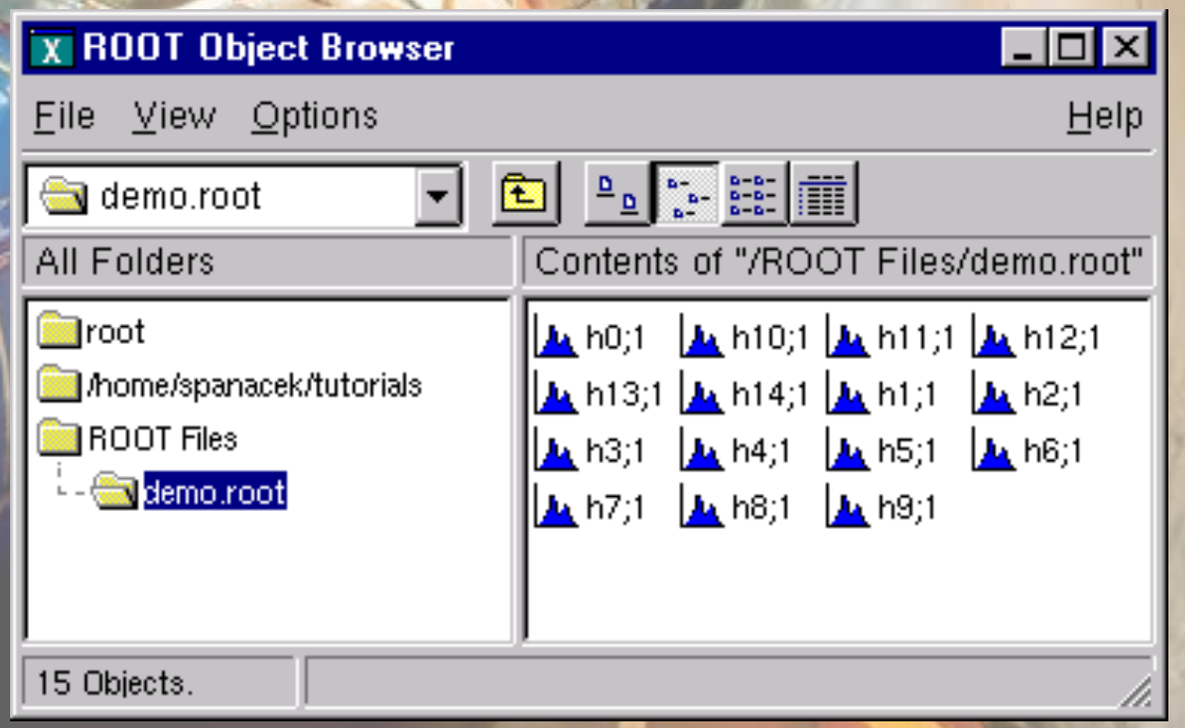


Genetic variants more often found in individuals with constrictions in small blood vessels (GWAS, Wikipedia)

Particle Detections as Data

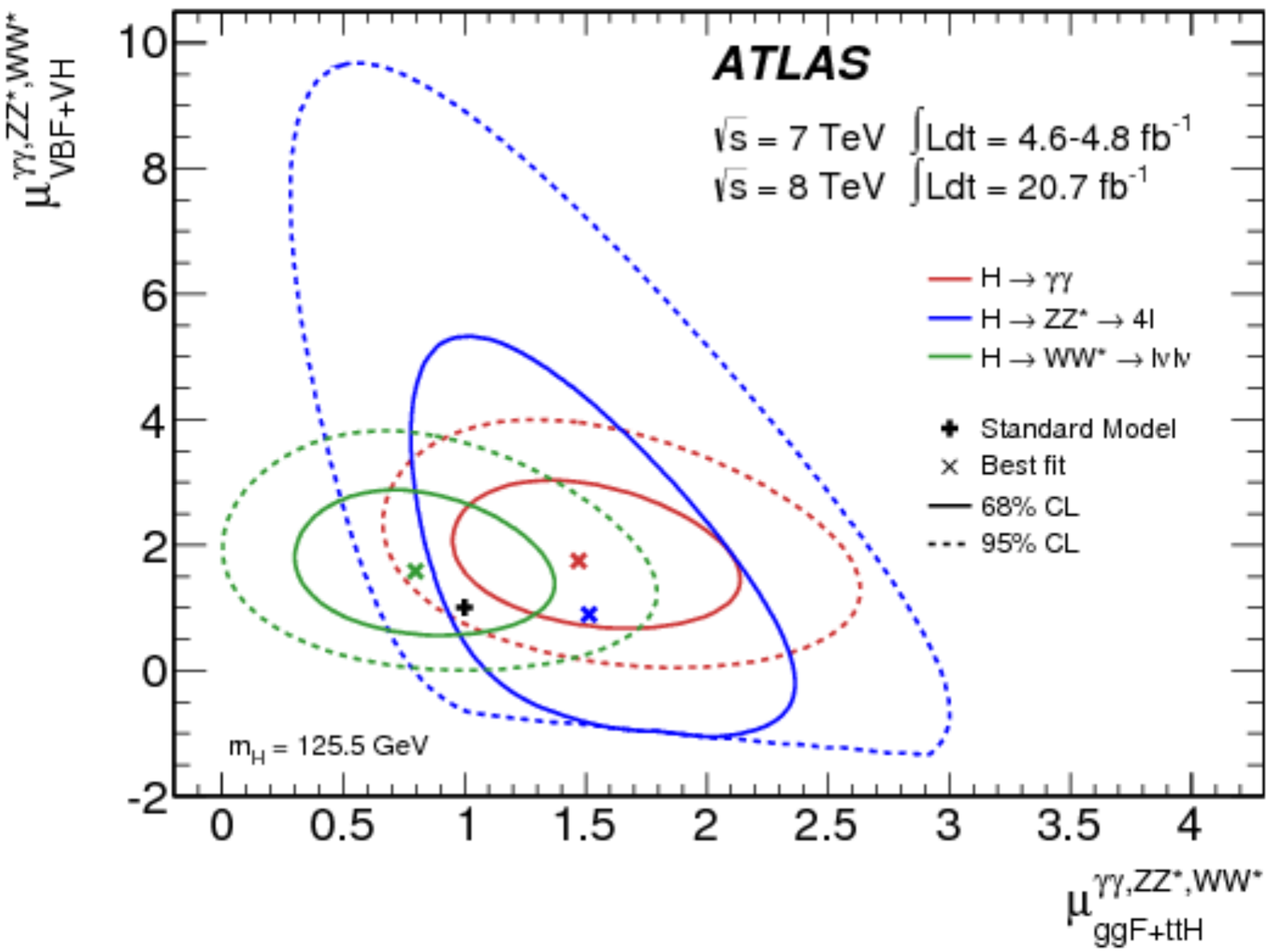
The Large Hadron Collider (HLC), CERN





Yes, there is a Higgs Boson!

(but is it the one we expected?)

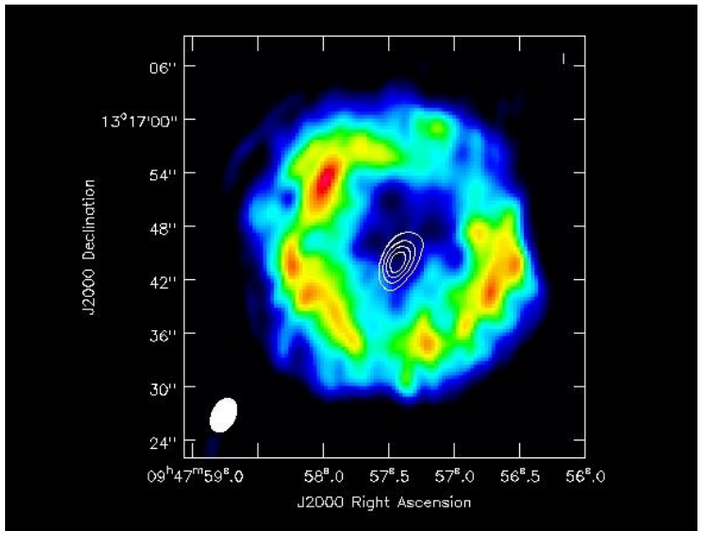


Likelihoods as result of comparing LHC observations with predictions from particle physics Standard Model

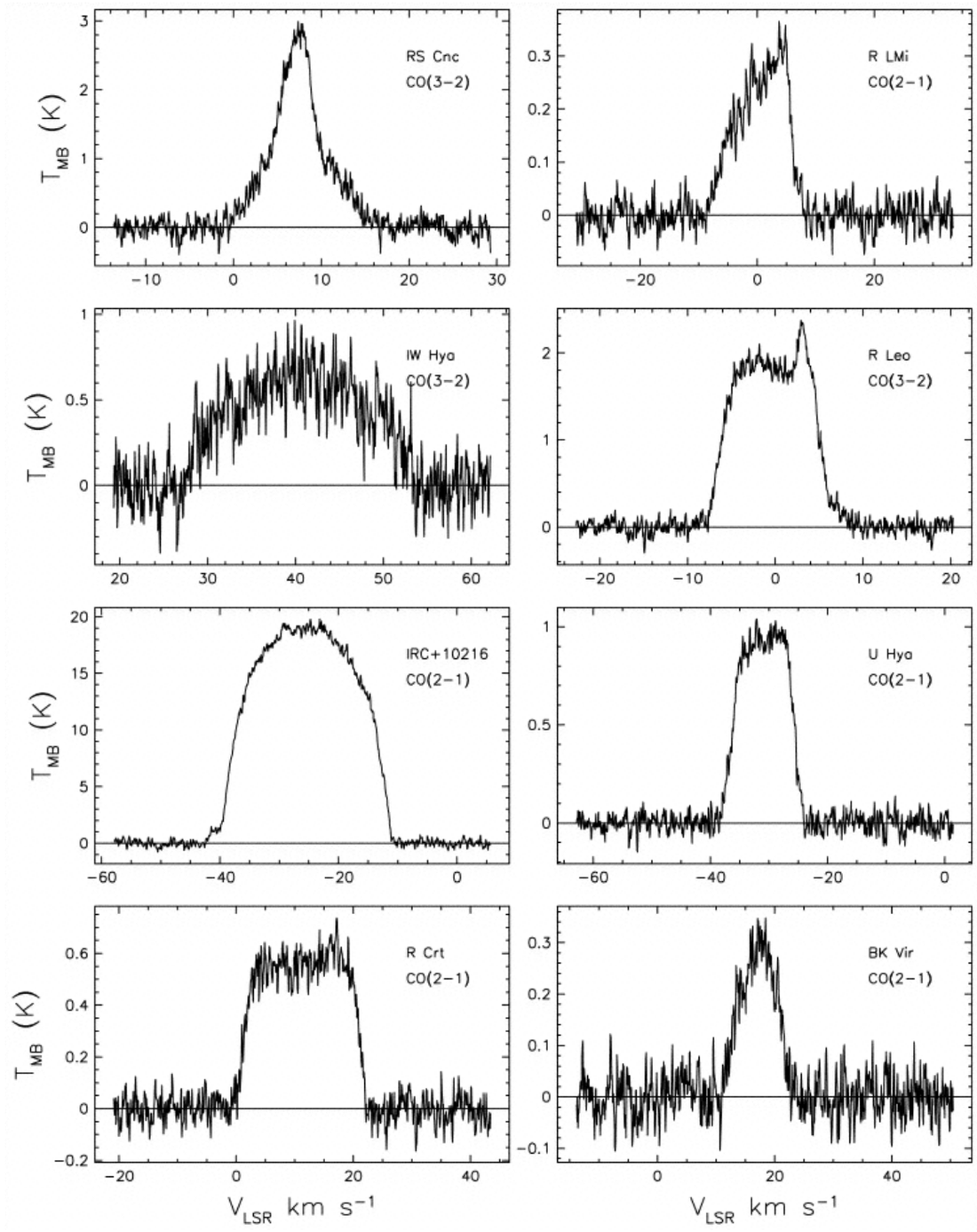
Measurements of Higgs boson production and couplings in diboson final states with the ATLAS detector at the LHC (Atlas Collaboration, 2013, with 2923 authors).

Acknowledgment Kyle Cranmer (NYU)

Radio Emissions as Data



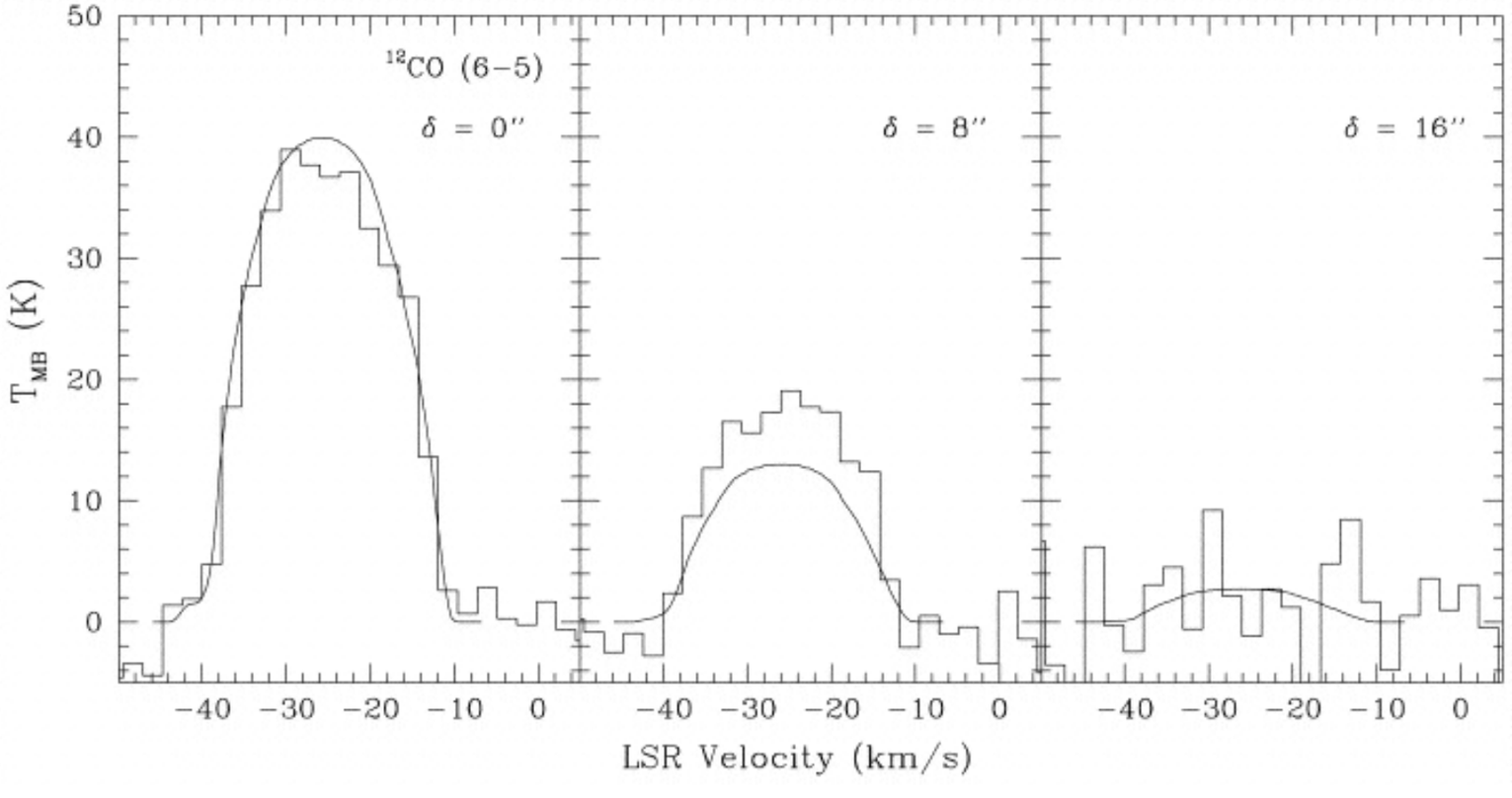
Caltech Submillimeter Observatory, California

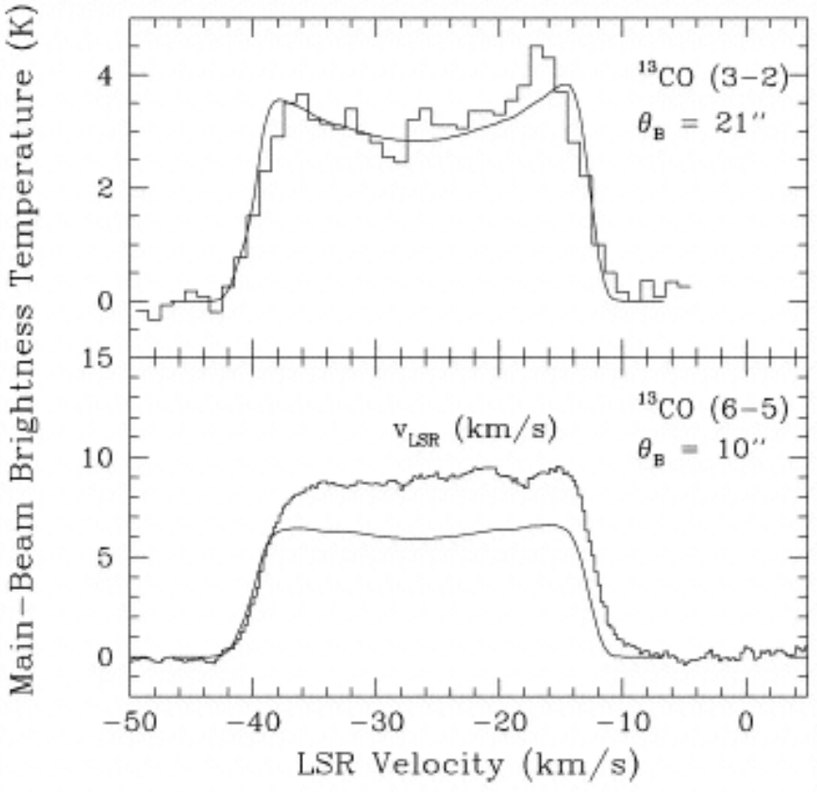


Temperature, Mass, and Distance derived from Stellar Molecular Lines

$$\frac{dT}{dr} = (2 - 2\gamma)\left(1 + 0.5\,\frac{d\ln v}{d\ln r}\right)\frac{T}{r} + \frac{\gamma - 1}{n_{\mathrm{H_2}}\,kv}\,(H - C)\,. \quad (11)$$

Monte Carlo simulations of radiative transfer compare observations with predictions from physical laws



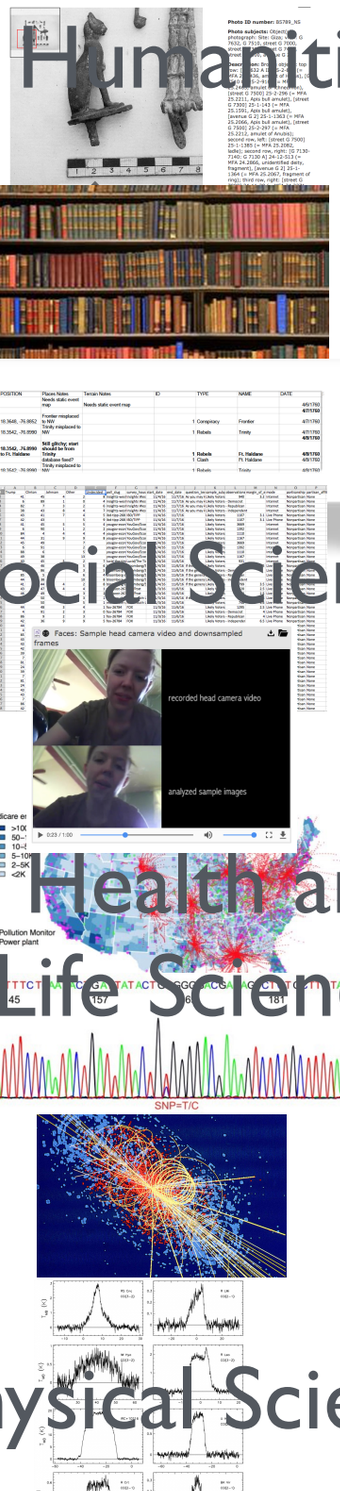


Crosas, Menten, Physical Parameters of the IRC+10216 Circumstellar Envelope (ApJ, 1997)

raw or primary data

order, transform, and analyze them

gain knowledge, make decisions



Humanities

Social Sciences

Health and Life Sciences

Physical Sciences

Catalog
Classify
Visualize
Quantify
Summarize
Geo reference
Inference
Missing data
Forecast
Causal Inference
Coding
Annotations
Associations
Likelihoods
Compare with theory

Learn about the whole from a part.

Tell a story.

Make a prediction.

Ultimately explain.

"Nullius in Verba"

(1665, the Royal Society motto and origin of Philosophical Transactions)

- **Replication***: Independent scientific experiments to validate findings
- **Reproducibility***: Calculation of quantitative results by others using original datasets and methods

(Stodden, Leisch, Peng, Implementing Reproducible Research, 2014)
* Replication and reproducibility definitions vary across disciplines

"Answering even a simple scientific question requires lots of choices that can shape the results"

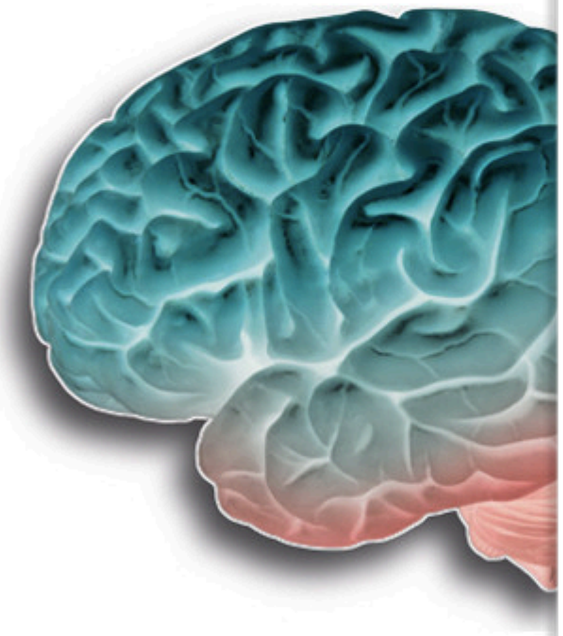
The Economist | World politics | Business & finance | Econ

Problems with scientific research

How science goes wrong

Scientific research has changed the world. Now it need

Oct 19th 2013 | From the print edition



A SIMPLE idea underpins science: "trust, but verify". Resu
challenge from experiment. That simple but powerful idea h
knowledge. Since its birth in the 17th century, modern scie
beyond recognition, and overwhelmingly for the better.

"When possible, make data, methods, and code open to verify"

INSIGHTS

Design principles for synthetic ecology p. 1425

Whacking hydrogen into metal p. 1429

SCIENTIFIC INTEGRITY

Self-correction in science at work

Improve incentives to support research integrity

By **Bruce Alberts,[1] Ralph J. Cicerone,[2] Stephen E. Fienberg,[3] Alexander Kamb,[4] Marcia McNutt,[5]* Robert M. Nerem,[6] Randy Schekman,[7] Richard Shiffrin,[8] Victoria Stodden,[9] Subra Suresh,[10] Maria T. Zuber,[11] Barbara Kline Pope,[12] Kathleen Hall Jamieson[13,14]**

Week after week, news outlets carry word of new scientific discoveries, but the media sometimes give suspect science equal play with substantive discoveries. Careful qualifications about what is known are lost in categorical headlines. Rare instances of misconduct or instances of irreproducibility are translated into concerns that science is broken. The October 2013 *Economist* headline proclaimed "Trouble at the lab: Scientists like to think of science as self-correcting. To an alarming degree, it is not" (*1*). Yet, that article is also rich with instances of science both policing itself, which is how the problems came to *The Economist*'s attention in the first place, and addressing discovered lapses and irreproducibility concerns. In light of such issues and efforts, the U.S. National Academy of Sciences (NAS) and the Annenberg Retreat at Sunnylands convened our group to examine ways to remove some of the current disincentives to high standards of integrity in science.

Like all human endeavors, science is imperfect. However, as Robert Merton noted more than half a century ago "the

activities of scie
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a result, as Popp
of the very few

POLICY

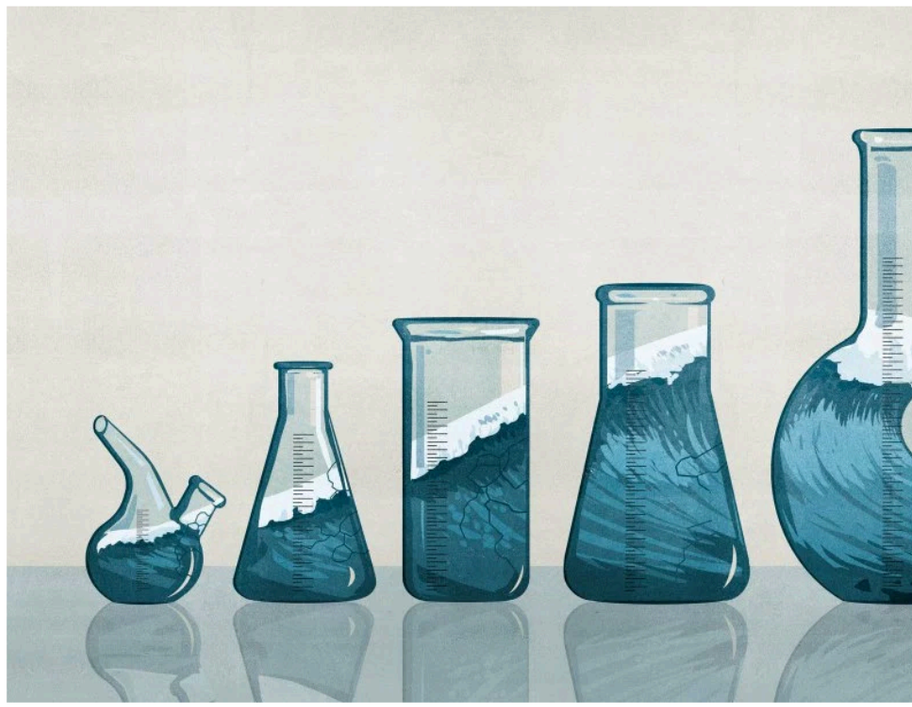
are s
fairly

(*3*). Instances in
address flaws in
of success, not f
onstrate the unc
nisms of science

Still, as in an
writ large does n
als. Although att
Wakefield study
between autism

"Science/research might be imperfect, but is self-correcting"

FiveThirtyEight Science



THE SCIENTIFIC METHOD | 7:00 AM | AUG 19, 2015

Science Isn't Broken

It's just a hell of a lot harder than we give it credit for.

By CHRISTIE ASCHWANDEN

Graphics by RITCHIE KING

If you follow the headlines, your confidence in science may have taken a hit lately.Peer review? More like self-review. An investigation in November uncovered a scam in which researchers were rubber-stamping their own work, circumventing peer review at five high-profile

"It's not unreliable, but more challenging that we give it credit for"

SCIENTIFIC DATA

Altmetric: 442 Views: 29,187 Citations: 32 More detail »

Comment | OPEN

The FAIR Guiding Principles for scientific data management and stewardship

Mark D. Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, Niklas Blomberg, Jan-Willem Boiten, Luiz Bonino da Silva Santos, Philip E. Bourne, Jildau Bouwman, Anthony J. Brookes, Tim Clark, Mercè Crosas, Ingrid Dillo, Olivier Dumon, Scott Edmunds, Chris T. Evelo, Richard Finkers, Alejandra Gonzalez-Beltran, Alasdair J.G. Gray, Paul Groth, Carole Goble, Jeffrey S. Grethe, Jaap Heringa, Peter A.C 't Hoen, Rob Hooft, Tobias Kuhn, Ruben Kok, Joost Kok, Scott J. Lusher, Maryann E. Martone, Albert Mons, Abel L. Packer, Bengt Persson, Philippe Rocca-Serra, Marco Roos, Rene van Schaik, Susanna-Assunta Sansone, Erik Schultes, Thierry Sengstag, Ted Slater, George Strawn, Morris A. Swertz, Mark Thompson, Johan van der Lei, Erik van Mulligen, Jan Velterop, Andra Waagmeester, Peter Wittenburg, Katherine Wolstencroft, Jun Zhao & Barend Mons - Show fewer authors

Findable
Accessible
Interoperable
Reusable

(data, code, methods)

Harvard Data Resources

Harvard Library Services and Archives

Harvard Dataverse (IQSS, Library, HUIT)

Research Computing Council

Harvard Data Group (OVPR)

VPAL Research Group

HMS Data Management group

Harvard Chan Bioinformatics Core

DARTH Arts and Humanities Computing

Center for Geographic Analysis

IQSS Data Science Services & Survey Research

HBS Research Computing Services

Services and Resources

Orchestra (HMS)

Odyssey (FAS)

RCE (IQSS)

HBS RC (HBS)

Computing & Storage Infrastructure

DataFest

Follow your sessions using SCHED

Tuesday, January 17

Data Concepts

Project Planning a Data Science Workflows

Wednesday, January 18

Processing and Analyzing Data*

Dissemination

Plan how to address your research question

Address it!

Share your results, data, code with others

*Does not include handling sensitive data

Don’t miss!

8:30am	Continental Breakfast	Computing Help Desk	
10:00am	Coffee Break	Archive-a-thon signup	Computing Help Desk
12:00pm	Lunch	Archive-a-thon signup	Computing Help Desk
5:30pm	Reception		

If you leave DataFest thinking "I need to learn more about X", we have already succeeded. (however, we welcome your feedback, using SCHED)

Harvard DataFest 2017

Schedule ▾ | Speakers | Attendees | About | Tickets | Search

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Nicole Alexander

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Sonia Barbosa

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Web Services Mgr, GIS Specia

Judson Harward
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Director, Arts&Humanities Research Computing

Amir Karger
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Associate Director of Research Computing

Radhika Khetani

Scott Lapinski
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Scholarly Communication & Research Librarian

JB

Jeff Blossom
Harvard University

Daina Bouquin
Harvard-Smithsonian Center for Astrophysics
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AC

Alex Caracuzzo
Baker Library, Harvard Business School
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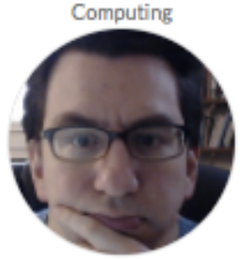
KC

Kareem Carr

BL

Ben Lewis



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